



LITE SOCIAL AND ENVIRONMENTAL BASELINE STUDY

COMMUNITY PERIPHERAL AREAS TO OPERATIONS IN UGANDA

JULY TO SEPTEMBER 2021

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1. Background

Since 2019 as a part of the Company's collaboration with WWF, WWF NGP and Forests Forward, NFC Uganda has adopted a landscape lens to our operations, that doesn't focus only on our impact on the Forest Management Units (FMUs) and Operational sites we manage but also the neighbouring Community tree plots and Community Associations we support and collaborate with. In 2019 we started the signing of Landscape Declarations with neighbouring communities, which committed us collectively to protecting the environment, while developing livelihoods.

In 2021, NFC began to engage with Communities adjacent to our operations about a potential carbon program at the landscape level. As part of that engagement, we committed to doing a lite assessment of the Environmental and Social baselines, to understand the key risks and priorities from the community perspective, to strengthen our landscape approach and reduce potential for doing harm while maximising potential for positive impact, in a transparent way.

A peripheral community is considered to be one which exists within 50km proximal to any of the three FMUs or the Pole Plant and has active direct engagement with the Company, through the New Forests Foundation and mutual projects such as employment, contract labour, Out-grower Association, Forest For Prosperity Association, Village Savings and Loans and Apiculture. The Districts covered include: Bugiri, Mubende, Kassanda, Mityana, Kiboga and Kyankwanzi.

2. Objectives

The Objective of this lite environmental and social baseline assessment was to understand the "AS-IS" scenarios, challenges, perspectives, and awareness of various risks, from the perspective of local government and communities. This would enable NFC to identify appropriate risk mitigation measures to strengthen the NFC carbon project and to use the results as a reference point going forward, for effective Monitoring and Evaluation, to see if scenarios improve or to act as an early warning in cases that conditions worsen over time.

The assessment was conducted by New Forest Foundation (NFF) staff, ensuring that all the main NFC communities and neighbours were covered. The survey was voluntary, with results aggregated and anonymized, following FPIC principles. The assessment included data relating to livelihoods, access to finance and PPI poverty related data, as these are priority community engagement areas for NFC, to strengthen our holistic approach to Uganda programmes. The assessment aimed to:

- Probe community and local government understanding of the current environmental and social risks
- Establish the level at which communities and government understand the need and role of environmental conservation.
- Understand the impact of environmental change on people's livelihoods.

- Gain a better understand on the existing conditions, weaknesses of riparian zone management and attitudes of the communities to wetland and riparian zone conservation.

3. Key Social Challenges

Communities identified a series of challenges they face, including the following:

- Youth unemployment.
- Deforestation.
- Biodiversity loss.
- Increasing encroachment on conservation areas.
- Access to adequate clean water.
- Climate change/unreliable weather.
- Lack of grazing grounds for cattle/goats.
- Inadequate capital.
- Gender-based violence.
- Access to health care.
- Lack of good cattle/goat breeds.
- Lack of clarity of environmental regulations

The communities were asked to rank these challenges in order of severity due to their local prevailing conditions. Thus, Youth unemployment ranked the highest problem, followed by poor access roads, challenges to access quality health care and clean water. Lack of land for cattle / goat grazing is also a significant challenge for the communities.

The community carbon project must help address these problems in the long term as they pose significant risk to promoting conservation and improving livelihoods within the landscape. NFC's own monitoring data and the Community Association project that performs risk management for the company, called Forests for Prosperity (FP) keeps scorecards which confirm the increasing pressure posed by grazing. Remote sensing data also verifies the impact of deforestation and increasing encroachment by communities on the environment.

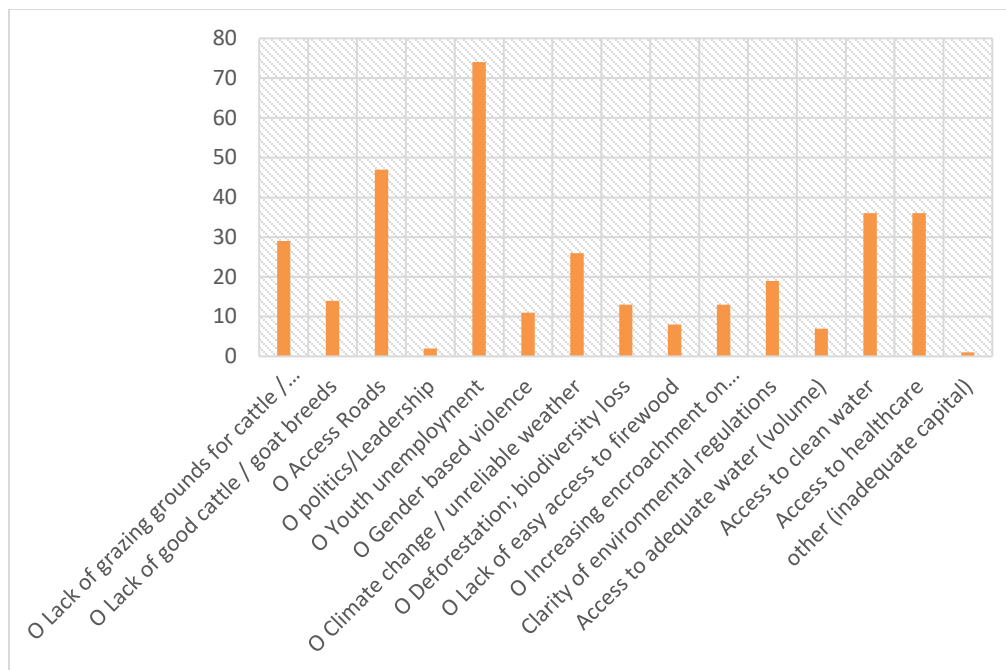


Figure 1: Responses from sampled peripheral communities: main challenges faced

Overall, there were 336 responses for communities peripheral to the NFC Forest Estates with the three most pressing problems they feel are hurting most, in descending order of severity are:

- Youth unemployment 74 (22%),
- Access to roads 47 (14%),
- Access to clean water 36 (10.7%)
- Healthcare 36 (10.7%)
- Lack of grazing grounds 29 (8.6%)
- Climate change/ unreliable climate 26 (7.7%)

4. Tree Planting Assessment

The tree growers were asked specific questions:

4.1 What is the main reason for planting trees?

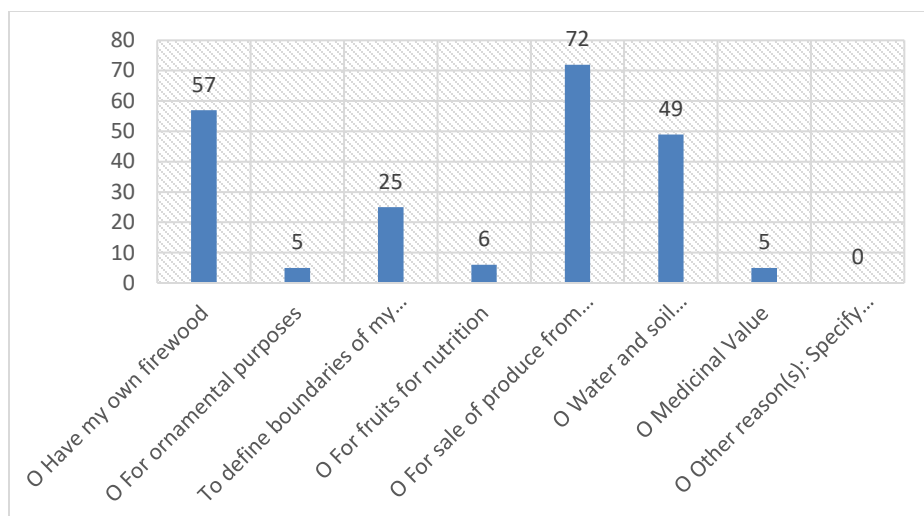


Figure 2: Reasons for planting trees

In total there were two hundred nineteen (219) responses to this question; Various reasons were given for tree planting by community members. The three most important reasons given for planting trees by communities were:

- For sale of produce from trees (72 respondents) 32.9%
- Have my own firewood (57 respondents) 26% and
- Water & soil conservation (49 respondents) 22.4%

The analysis results are understandable. Populations are doubling every 20 years in Uganda so daily household income for communities is their main concern. Communities struggle to find income to solve daily domestic demand of ordinary items of food, salt, sugar, medicines, education, burial expenses etc. This finding on the challenge of households & Outgrowers incomes and poverty is consistent with findings from previous studies undertaken by NFC and its partners including Out-grower pain points analysis, SDM case study (conducted with funding from IDH) and FP baselines (conducted with funding from FMO). Any community centered strategy therefore must be designed to solve this increasing and persistent problem.

Other top reasons for planting trees are soil conservation firewood availability. Most rural households don't have access to electricity and rely on firewood for cooking at least two meals a day. The need for firewood is an indicator of the importance of encouraging tree growing for multiple purposes by households e.g., food security and firewood in addition to commercial species.

Medicinal, ornamental values and boundary definition/marketing seem not to be of high value and/or concern to communities at present.

The present circumstances of relative wood availability will not continue given the increasing population of Uganda. The situation is not sustainable with deforestation and charcoal businesses growing. Uganda's current population growth rate of 3.7% (2018), 4.96 births per woman (2018) and youth population at 75%, is a major concern for wood resources given that over 95% of

Uganda's population (and probably 100% of communities mentioned herein) depend on firewood for cooking daily.

4.2 Would you plant more trees if seedlings and support are available?

This was a Yes or No Question. Ninety-eight (98) of 106 Out-growers sampled (92%) would definitely plant trees if seedlings and support were provided. This affirms the Company's investment in the Out-grower program and the possibility to expand it, to help solve livelihoods and climate challenges facing communities.

4.3 What are the biggest challenges with tree growing?

The responses were as below:

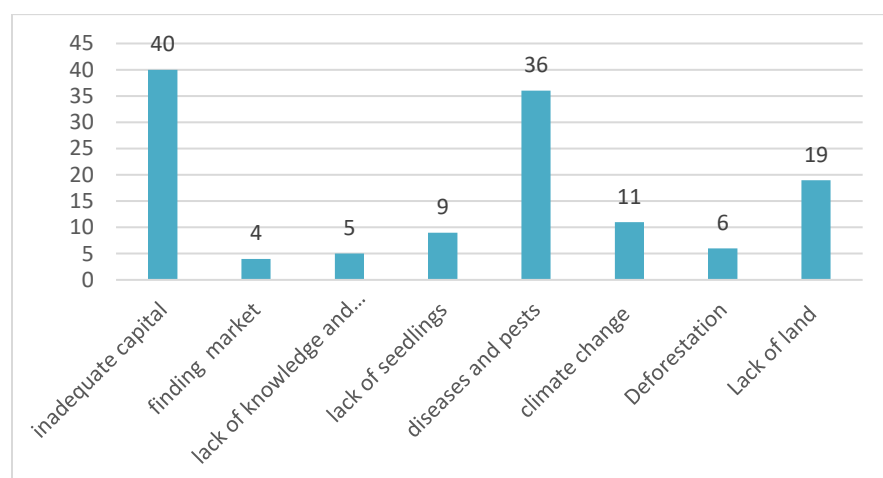


Figure 3: Challenges with tree growing

A total of one hundred thirty (130) responses was recorded for which the main challenges identified hampering tree growing are:

- inadequate capital (30.8%),
- diseases, and pests (27.7%) and
- lack of land (14.6%).

The lack of seedlings remained significant challenges to tree growing. Suboptimal seedling quality may be linked to diseases and pests as well as survival issues. Without money/capital/external support not many farmers will be able to grow trees. The analysis also shows that farmers are tied to primary challenges, that of acquiring the basics for establishing a tree farm. So, the other challenges could not be experienced and/or assessed i.e., one needs to have established a tree garden before s/he can assess issues regarding climate, pests, diseases, technical knowledge & skills or market. Trees take 10-15 years to reach maturity and many of the NFC Out-growers have only been growing trees since 2015, so they have not yet needed to find a market for their timber.

4.4. What can be done to mitigate/help overcome the challenges with Tree Growing?

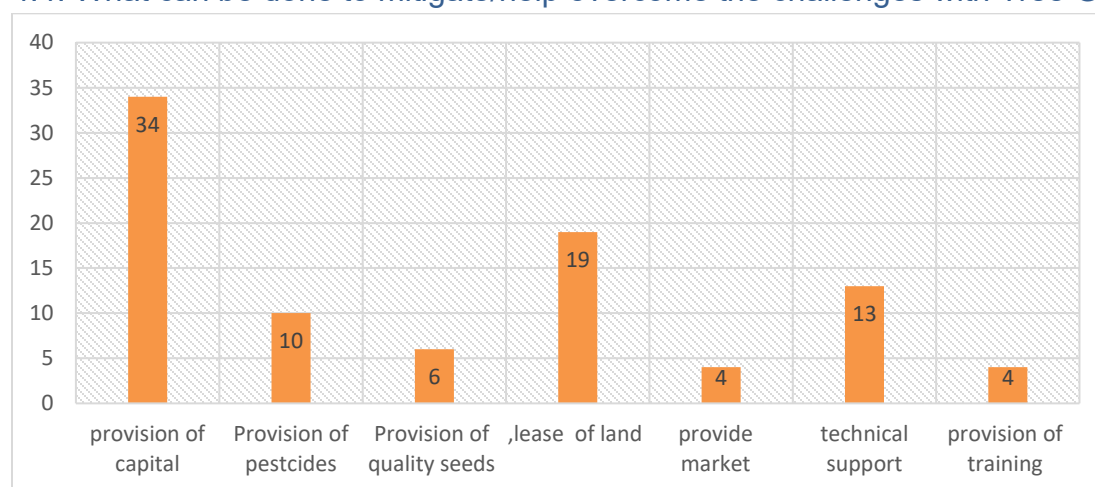


Figure 4: Mitigation of challenges

A total of 90 responses were recorded where various interventions were suggested to mitigate the challenges to tree growing by communities. The key interventions identified are:

- provision of capital (37.8%) probably linked to the stated need for inputs including provision of quality seedlings and pesticides);
- need to lease/access land (21.1%); this could also impact youth and women participation in Tree Growing, as land is typically carried down the male lineage in Uganda) and
- technical support to Out-growers (14.4%).

4.5 Is deforestation a problem?

This was a “Yes or No” Question. Sixty-three (63) of 106 respondents (59%) think deforestation is a problem that is linked to the climate and environmental degradation.

4.6 Is climate change a problem?

This was a “Yes or No” Question. Fifty-two (52) of 106 respondents (49%) think climate change is a problem.

5. Biodiversity Assessment

5.1 What are the most common species in your area?

The participants in the survey were asked to list biodiversity in their area. This is the collated response from 63 respondents

Animals - cattle, goats, rabbits, monkeys, monitor lizards, squirrels, snakes, rats, pigs.

Birds – hens, weaver birds, pigeon, guinea fowl, eagles, egrets, doves

Fish – tilapia, Nile perch, mud fish, silver & lung fish (please note this is for Kirinya and Namwasa only. The other sites report n/a)

Insects – bees, grasshopper, butterflies, tsetse flies, house flies. termites, mosquitoes, wasps, ants

Other common creatures – Reptiles,

It was not possible to get pictures at the time of doing the survey, but this is noted as a needed activity going forward.

It was also noted that trees, plants, flowers were not considered / listed under biodiversity, indicating that increased awareness in this area is also needed.

5.2 Do they participate in the protection of these species?

The surveyed participants were asked to elaborate if they had answered Yes to participating in protection of species

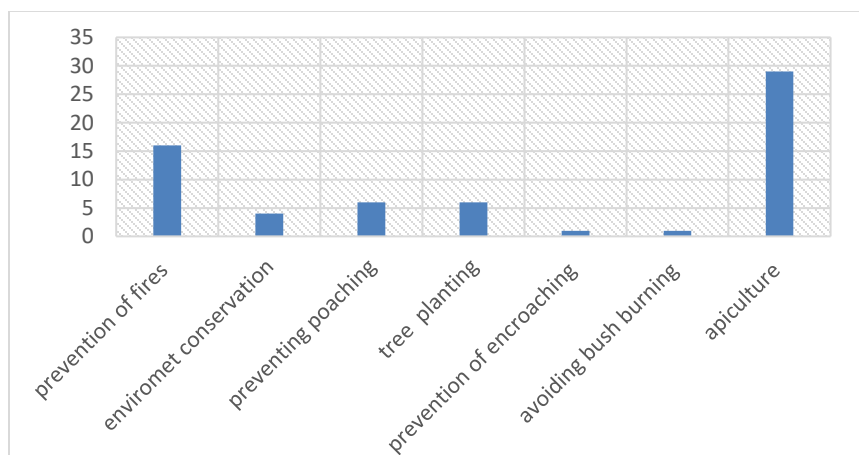


Figure 5: Participation in conservation

Sixty-three (63) responses were captured in total where the main methods are

- engaging/encouraging apiculture (46%),
- preventing fires/bush burning (25.4%),
- tree planting, and preventing poaching, both at (9.5%).

Focus on fire prevention, tree planting and encroachment portends well for conservation efforts and the carbon project.

5.3 How important is conservation to you?

In this question, respondents were asked to rate the importance of conservation using a score between 1 and 5, where: 5 is very important and 1 is Not important at all)

All 94 respondents (100%) reported that conservation is very important to them. This is likely because they are aware of changing rain patterns and the reduced availability of open areas, as populations increase. The amount of time it takes to collect firewood has increased and the rainy seasons have anecdotally become shorter, but heavier. NFC and WWF landscape Stewardship meetings have also raised awareness on the need for conservation with communities.

6. Cultural Assessment

6.1 Are there any indigenous* people in the area?

This was a “Yes or No” Question. The definition of “indigenous people” was taken from UNDRIP ILO Convention 169 and translated into local languages to ensure common understanding.

All 106 respondents (100%) across all three plantations and the pole plant communities confirmed that there are no indigenous peoples in their communities. This is consistent with the information the Company had established at the time of establishing the operational sites.

6.2 Are there areas of cultural significance?

This was a “Yes or No” Question. For Kirinya 25 people (100%) the response was “Yes”. For Namwasa and the Pole Plant some “yes” responses were recorded. For Luwunga there were also a few “yes” responses, but these need further investigation.

The Kirinya areas relate to a rock that is known for its beauty. The Namwasa areas relate to a cultural shrine that is important to the Baganda. The Pole Plant areas relate to graves on the pole plant yard. These results indicate a need to create awareness and even ensure an FPIC (Free, Prior and Informed Consent) process at Luwunga on the NFC ASI prescription which ensures a proper process to identify, guarantees full access and no charges to all ASI (Areas of Significant interest)

6.2.1 If yes, are they mapped?

This was a “Yes or No” Question. For Kirinya and Namwasa these areas were already mapped. For the Pole Plant the process of mapping is underway. For Luwunga; The information is unclear (from third parties) and it requires more investigation.

6.2.2 If yes, who owns / protects these areas?

- All 25 respondents (100%) at Kirinya believe the Community look after the area
- 20 at Namwasa (80%) provided the name of the caretaker Nalongo Mujinja from the Community
- 30 at Pole Plant (100%) responded that the Community look after the area
- No one knows who currently owns/protects the Luwunga site. It is still “hear-say” that needs to be further investigated.

These results indicate a need to create awareness and even ensure an FPIC process on the NFC ASI prescription which guarantees full access and no charges to all ASI.

6.2.3 If yes, how is access to such areas guaranteed?

- 100% of Kirinya respondents (25) think there is free access
- 20 (80%) at Namwasa say they seek permission for access, but don't say from who
- 30 (100%) at Pole Plant think community manage access
- No response from Luwunga.

These results indicate a need to create awareness and even ensure an FPIC process on the NFC ASI prescription which guarantees full access and no charges to all ASI.

7. Financial Inclusion

7.1 Source of capital for households

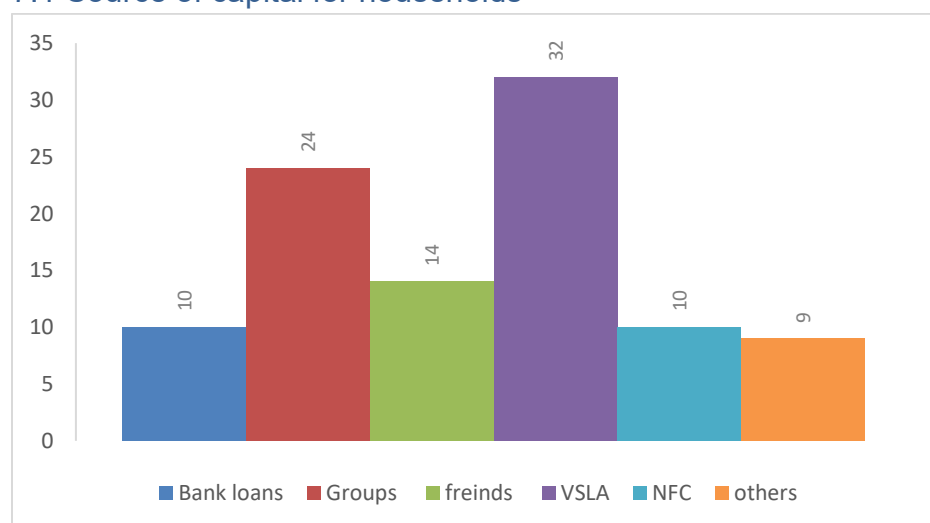


Figure 6: Source of capital

Ninety-nine (99) households were sampled. The majority of households still depend on VSLA (Village Savings and Loan Associations) savings at (32) 32.3% to bail themselves out of financial difficulties. A lesser number (24) 24.2% make use of Community groups (CA and OA) and (14) 14.1% rely on borrowing from friends. Though loans from commercial banks are increasingly being sought, their uptake is still low (10), and comparable to community members seeking financial assistance from sources other than those mentioned herein (see graph above).

7.2 Access to Credit Facilities

This question delved further into which organizations were relied upon for credit facilities

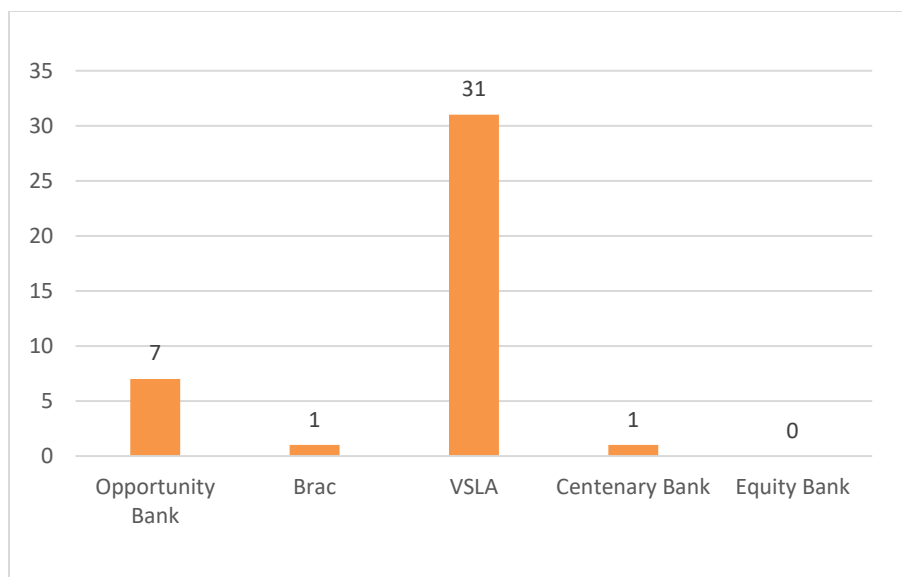


Figure 7: Organisations that offer loans

A total of 43 responses was recorded at this section. There are two major sources of credit the households rely on for credit: Community Associations – VSLA (31) 69.8% i.e., internal savings and Opportunity Bank is slowly gaining ground in this regard (7) 16.3%. This is especially good for people who want to borrow money for small business ventures within the communities. It is worth noting that effort by NFC to court Opportunity Bank into business with communities is slowly paying-off, evidence being that other banks are barely known because they are not operational in these areas. A total of 43 respondents were comfortable to disclose this information. Opportunity Bank, BRAC, Centenary Bank, Equity Bank are formal financial institutions licensed by the Central Bank of Uganda to operate.

7.3 What was the Total Loan Amount in Uganda Shillings that your household obtained in the last year 2020?

60 respondents report borrowing more than UGX 1M (USD280) while the remaining borrowed less than UGX 1M (USD20).

7.4 What types of activities did loans contribute toward?

Graph 8: Use of loans

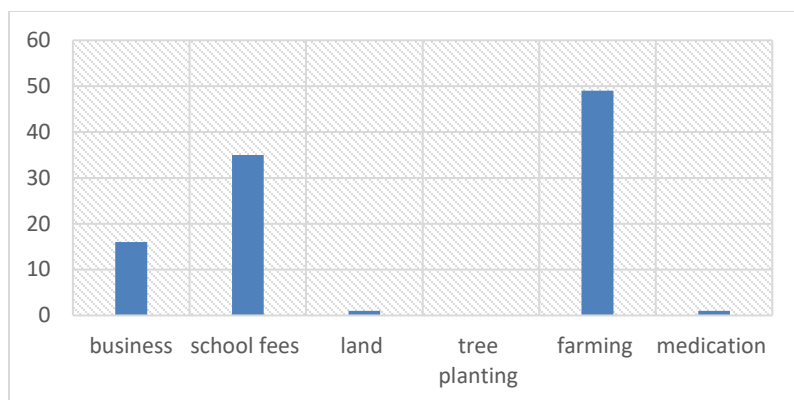


Figure 8: Use of loan

A total of one hundred and two responses were recorded to cover loan purpose. Borrowed funds are mainly used to pay for:

- farming (48) 47.1%
- children school fees (34) 33.3%,
- business (16) 15.7%.

It can be noted that when a family is in dire straits and schools require that fee arrears be cleared, the only option is to borrow money. However, money is also borrowed for farming activities and small business investments.

7.5 What was the impact of the loans on the wellbeing of your household?

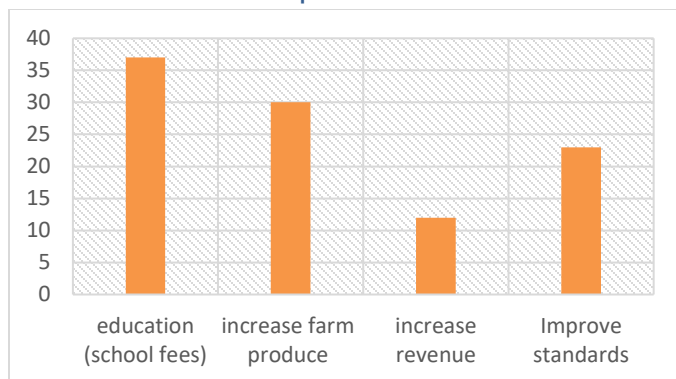


Figure 9: Impact of loans

Out of one hundred (one hundred) total respondents (36) 36.6% report that loans help households meet school fees and so reduce anxiety/stress; increase farm produce/production (29) 29%; improve living standards (23) 23% and generally increase household income (12) 12%. This finding is consistent with the stated intended purpose of borrowing.

7.6 Availability of insurance

Communities were asked if they were aware of any Insurance Companies working in their area (in case of any unforeseen emergency?). It was a “Yes or No” question. 63 out of 100 of respondents (63%) had not heard of any insurance company.

7.6.1 What insurance companies are working in your area?

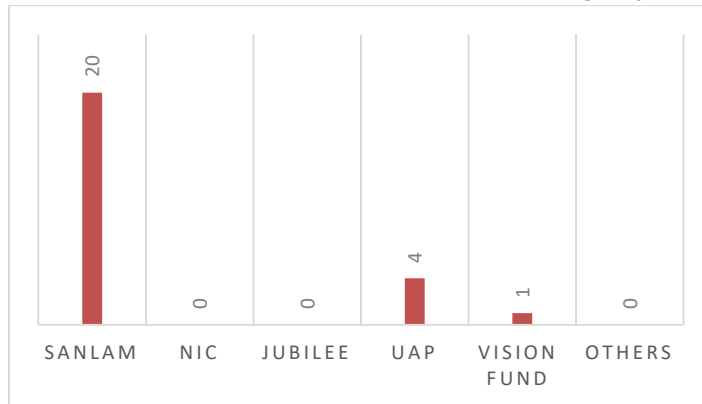


Figure 10: Insurance companies

Despite Uganda having 18 Insurance Companies, 12 Non-Life and 6 Life Insurance Companies in Uganda, only three are visible in the communities neighboring NFC Forest Estates (see graph above): Sanlam (20) 80%, Vision Fund (1) 4% and UAP (4) 16%. A total of 25 respondents proved awareness about insurance Companies working in their areas.

There is a lot of work still to be done in this area since insurance services are and will be necessary for a long time to come. Even those that have used insurance services, they mainly do under compulsory circumstances e.g., obtaining Third Party insurance, a compulsory policy required by government for all vehicles. It is thus most likely that these were commercial motorcycle riders (commonly called '*boda-boda*') who had to compulsorily acquire this type of insurance so that they would not be apprehended by Law Enforcement Officers.

8. Health and Sanitation

8.1 Are livestock and/or chicken kept in the same house used for sleeping?

It is very encouraging that out of 103 respondents (102) 99% do not keep livestock and or chicken in the house used for sleeping. The reason is that this helps control/avoid disease in the homestead; and separate shelters are kept livestock and or chicken.

8.2 Does the home have plumbing?

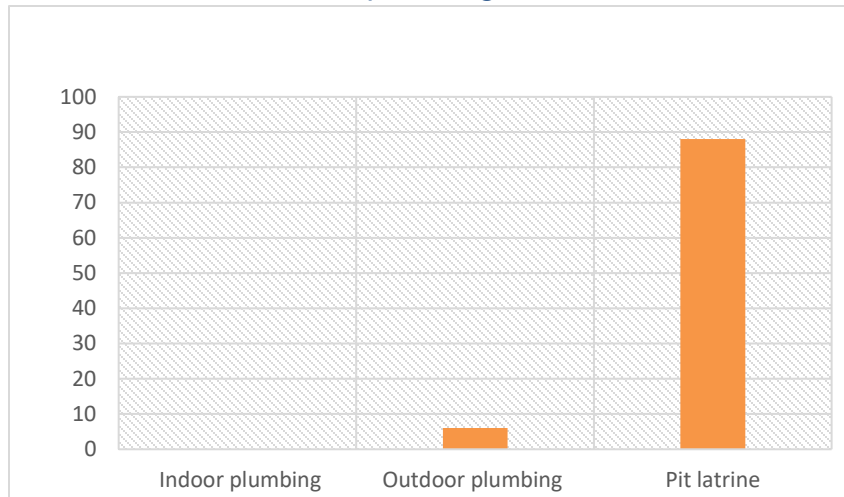


Figure 11: Health and sanitation

The plumbing arena clearly shows the social level of the NFC communities. Absence of indoor plumbing and 88 (93.6%) availability of pit latrines out of 94 responses indicates a typical rural African village setting. Presence of outdoor plumbing at 6 (6.4%) shows presence of small trading centers where the Town Council or urban authorities have extended piped water.

8.3 Do you know government laws about waterways?

This was a “Yes or No” Question. About half the respondents (54) 51.4% out of 105 respondents know about government laws about waterways while 51 (48.6%) don't.

8.4 Can you use the water ways for agricultural production on your land?

This was a “Yes or No” Question. Only about one third (38) 35.8% out of 106 know that you can use waterways for agricultural production on your land while the majority (68) 64.2% don't know this.

8.4.1 If yes, do you follow these rules?

This was a “Yes or No” Question. Most respondents (33) 71.7% report following the rules governing water ways while 13 (28.3%) out of 46 total responses report not following the rules.

8.5 How important is it to protect waterways?

Respondents were asked to rank their answers between 1 and 5, where 5 is Very important and 1 is Not important at all. Fifteen (15) of 18 respondents (83%) who answered this question think that it is very important while the rest think it is Important to protect waterways.

8.6 Do you access waterways on your land or nearby land for home use?

This was a “Yes or No” Question. 48 (45.7%) out of 105 can access waterways on their land or nearby land for home use while 57 (54.3%) don’t.

8.6.1 If yes, how do you know the water is safe to drink?

There were 54 responses in total: Fewer people responded to this question which might be attributed to limited technical knowledge on waterways by the sampled communities The water is safe to drink because:

- It flows – 6 (11.1%)
- Its protected – 8 (14.8%)
- Never got sick – 11(20.4)
- Water quality tested by government -30 (55.6%)

It’s important to note that the respondents who reported that water is tested by government were all from the Pole Plant area which is covered by the NWSC. The other interesting response was linking water quality to health by 20% of respondents.

8.7 Do you access waterways on your land or nearby land for agriculture use?

This was a “Yes or No” Question. Out of 98 total responses 49 (50%) respondents report having access to waterways on their land or nearby land for agricultural use while the other half (50%) does not have access.

8.8 Do you use fertilizer/organic manure on your land?

This was a “Yes or No” Question. Seventy-one (71) out of 82 respondents replied in the affirmative (86%).

8.8.1 What type of fertilizer is used?

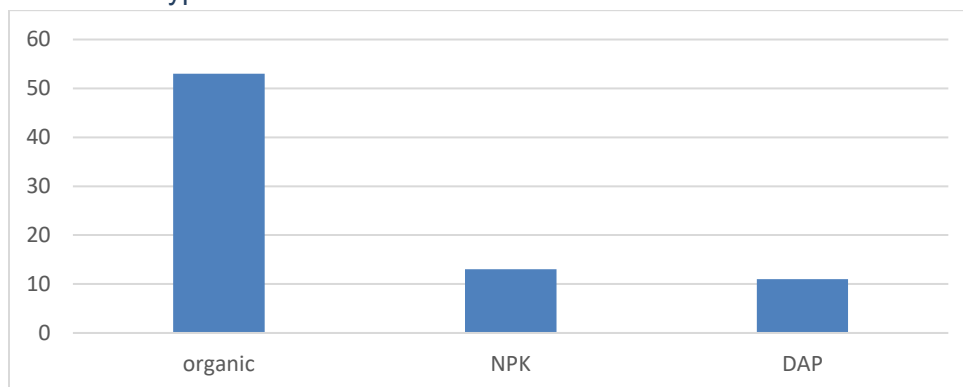


Figure 12: Types of fertilisers

The most common fertilisers used by respondents are: Organic 53 (68.8%), NPK 13 (16.9%), and DAP11 (14.3%) out of 77 total responses.

8.8.2 How much is applied per acre?

Total responses were 64.

- Above 50kg/acre -23 (35.9%)
- Below 50kg/acre – 11 (17.2%)
- 900kg/acre – 30 (46.9%)

8.9 Do you use pesticides?

Seventy-eight (78) of 81 respondents confirmed they use pesticides (96%).

8.9.1 What type of pesticides are used?

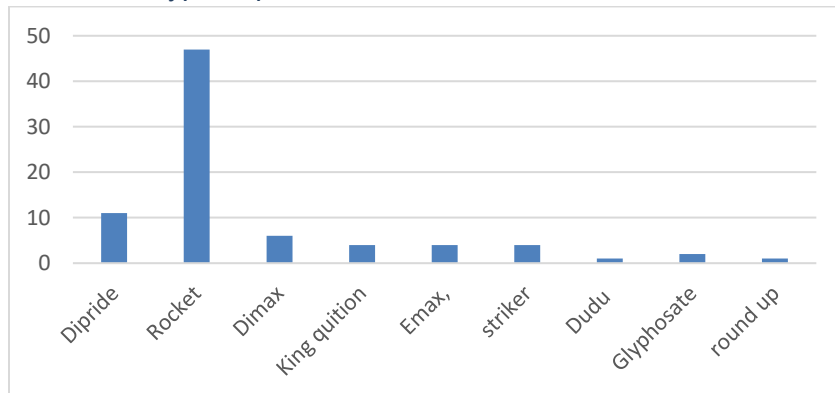


Figure 13: Types of pesticide used

There were 80 responses:

- Dipride – 11 (13.8%)
- Rocket 47(58.8%)
- Dimax 6 (7.5%)
- King gution, Emax, Striker – 4 (5%) each
- Glyphosate/Round Up 2 (2.5%) and

8.9.2 How much is applied per acre?

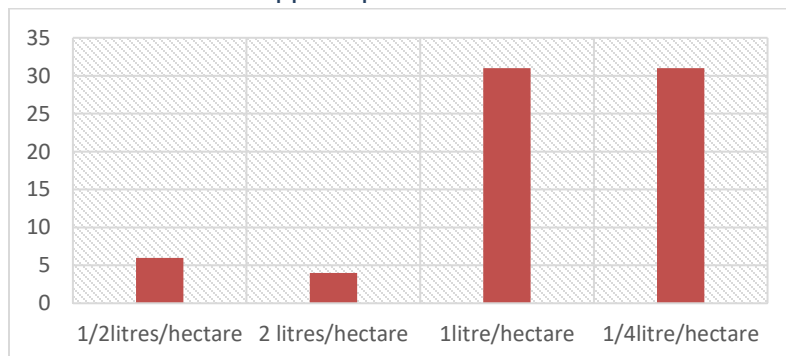


Figure 14: Amount of pesticide applied

Total responses:72

- 0.25 litres/ha – 31 (43.1%)
- 0.5 litres/ha – 6 (8.3%)
- 1. litres/ha – 31 (43.1%)
- 2.0 litres/ha – 4 (5.6%)

8.10 Do you understand what conservation areas are? If yes, what are the values of protecting conservation areas.

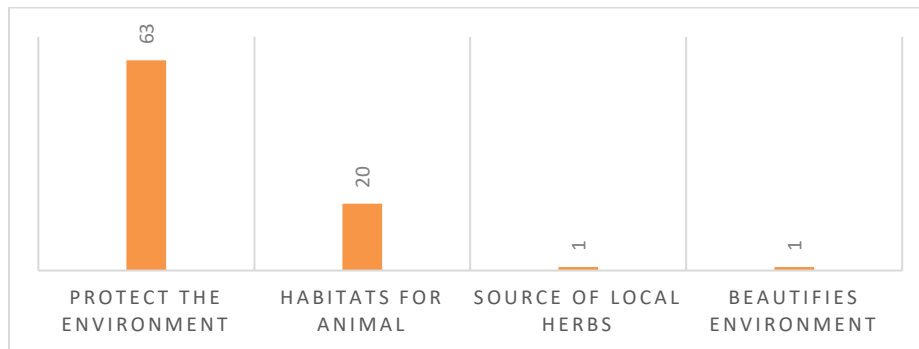


Figure 15: Understanding of conservation value

There were 85 responses:

- Protect the environment – 63 (74.1%)
- Habitats for animals – 20 (23.5%)
- Source of local herbs – 1 (1.2%)
- Beautiful environment – 1 (1.2%)

8.11 How close is your source of safe water point?

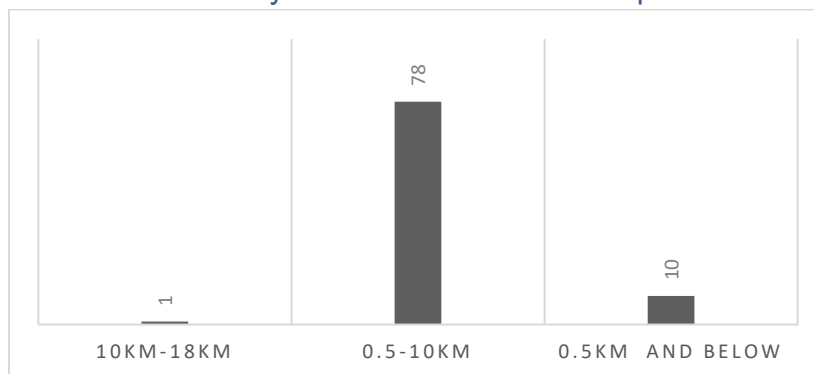


Figure 16: Proximity of water source

- 0.5km and below – 10

- 0.5km -10km – 78
- 10km-18km – 1

8.12 Is your domestic water source protected?

This was a Yes or No question with 31 (72%) responding affirmatively and 12 negatively.

8.13 Do you have sufficient rainfall for your crops every season? If not, what do you think are the main causes?

This was a Yes or No question with 20 (27%) responding affirmatively and 55 negatively.

Respondents reported that rainfall is insufficient for their crops every season giving deforestation as the main cause.

9. District Local Governments: Bugiri, Kassanda, Kiboga and Mityana

Each head of department was interviewed and provided the perspectives included in this report under the different departments

A. Department of Environment and Natural resources

There were 4 respondents:

- a. **What are the main challenges reported by the District Local Government? (Rank in order of decreasing severity i.e., from most to least severe in the list).**

Top Key issues reported include: Forest destruction, Land degradation, Wetland encroachment, Water pollution/water flow challenges and to some extent increased land fragmentation.

- b. **Land ownership/ use type?**

The tenures as per the constitution: customary – Kibanja, Family, Clan land; Mailo, freehold; leasehold, Kibanja interest on Mailo.

- c. **What is the average landholding size of a homestead**

Average seems to be 1 acre

- d. **Typical land use breakdown?**

Predominantly food production with some land left for homestead, grazing and growing trees.

- e. **Is there a significant trend in land use away from farming to woodlots? (Yes/ no Or rank on a scale of 1-5, where 1 = low likelihood and 5 = high likelihood?)**

Low Likelihood

- f. **Is there a significant trend in land use away from farming to other activities including migration to towns/trading centres? (Yes/ no Or rank on a scale of 1-5, where 1 = low likelihood and 5 = high likelihood?)**

Low Likelihood

- g. **Is there Displacement of damaging activities to outside the project area (Yes/ no Or rank on a scale of 1-5, where 1 = low likelihood and 5 = high likelihood?)**

Low Likelihood

- h. **Do you have any recent Environmental or Social reports or statistics for the district that could be shared?**

Copies of District Development Plans availed

- i. **Are there any targets and progress plans for these reports or statistics for the district?**

Yes

B. Department of Production

There were 4 respondents:

- j. **What are the main products being produced in this district?**

Maize, Coffee, Beans, Bananas, Fish, honey, rice, milk, Livestock products, cassava, charcoal, electricity poles

- k. **What are the main challenges reported by the District Local Government? (Rank in order of decreasing severity i.e., from most to least severe in the list).**

Lack of knowledge on soil improvement techniques and Lack of access to markets are the top two followed by Lack of farm inputs and low crop yields as the next two.

- l. **What is the average Livelihoods breakdown in the sampled area? -----% cash crop, ---% Trees, and -----% other (e.g. 40% cash crops + 30% trees + 30% apiculture)?**

Generally, crops over 60% and trees around 15

C. Department of Trade, Industry and Local Economic Development

- m. **What are the main economic opportunities in the district?**

Agriculture & livestock farming, Horticulture, Beef & Dairy, Apiculture, Tourism, Finance, Commercial agriculture, Trading, Agro-forestry, Tourism, Sandmining & Stone mining

- n. **What are the main economic opportunities in the District Local Government? (Rank in order of decreasing severity i.e., from most to least severe in the list).**

Predominant: Lack of business infrastructure e.g., roads, internet, legal and banking?
Fewer business opportunities followed by Lack of starter capital and Lack of business infrastructure e.g., roads, internet, legal and banking.

D. Department District Community Development

- o. **What are the main community development projects in this district?**

National Agricultural Advisory Development Services (NAADS)/ Operation Wealth Creation (OWC) – Central Government Programs implemented through Local Governments.
FISHERIES; VSLA & Function Adult Program (FAP)

- p. **What are the main challenges reported by the District Local Government? (Rank in order of decreasing severity i.e. from most to least severe in the list).**

Key: Increasing early marriages, early pregnancies; increases theft/crime and access to health care

E. Department of Education

- q. **What are the main focus areas of the district?**

Universal Primary Education (UPE), provide education, addressing poverty & domestic violence, Quality social service

- r. **What are the main challenges reported by the District Local Government? (Rank in order of decreasing severity i.e. from most to least severe in the list).**

Key issues: Lack of pedagogical and scholastic materials; Increased rate of school dropouts, Long distances of commuting from School-Home, Lack of school infrastructure

- s. **What % complete primary school?**

Ranges between 60% to 87% across the 4 districts.

- t. **What % complete High school?**

Ranges between 30% to 69% across the 4 districts.

- u. **What % complete tertiary education?**

Ranges between 10% to 60% across the 4 districts.

Summary of the responses per district

We interviewed department heads from 4 districts, so each department provided 4 responses. The top issues listed are the ones that are in the top four for the districts

	KIRINYA	NAMWASA	LUWUNGA	POLE PLANT
A: Department of Environment and Natural resources				
What are the main challenges reported by the District Local Government? (Rank in order of decreasing severity i.e. from most to least severe in the list).				
	Water pollution / water flow changes	Increased Land fragmentation	Wetland Encroachment	Forest Destruction
	Forest Destruction	Water pollution / water flow changes	Land degradation	Poverty
	Land degradation	Forest Destruction	Increased Land fragmentation	
	Wetland Encroachment	Land degradation	Wetland Encroachment	
	Climate Change	Climate Change	Forest Destruction	
	Poverty	Wetland Encroachment	Climate Change	
	Increased Land fragmentation	Poverty	Water pollution / water flow changes	
Land ownership/ use type?				
	Customary	Freehold Title	Freehold Title	Mailo Title
	Family/Clan Land	Family/Clan Land	Mailo Title	Kibanja interest on Mailo Title
			Freehold Title	Family/Clan Land
			Kibanja interest on Mailo Title	
What is the average landholding size of a homestead				
	1 acre	1/2 acre	3.5 acres	1 acre
Typical land use breakdown?				

Trees	10%	15%	15%	20%
homestead	10%	15%	10%	25%
Food production	70%	50%	60%	21%
Grazing/ pasture lands	10%	20%	15%	10%
	Is there a significant trend in land use away from farming to woodlots? (Yes/ no Or rank on a scale of 1-5, where 1 = low likelihood and 5 = high likelihood?)			
	Yes=1	yes=1	yes=1	yes=1
	Is there a significant trend in land use away from farming to other activities including migration to towns/trading centres? (Yes/ no Or rank on a scale of 1-5, where 1 = low likelihood and 5 = high likelihood?)			
	Yes=1	yes=1	yes=1	yes=1,
	Is there Displacement of damaging activities to outside the project area (Yes/ no Or rank on a scale of 1-5, where 1 = low likelihood and 5 = high likelihood?)			
	No=0	yes=1	No=0,	No=0,
	Do you have any recent Environmental or Social reports or statistics for the district that could be shared?			
	Yes=1	No=1	yes=0, No=0	Yes=1,
	Are there any targets and progress plans for these reports or statistics for the district?			
	Yes=1	No=1	yes=1	yes=1
B. Department of Production				
	What are the main products being produced in this district?			
	Fish, honey, maize, rice, coffee, milk	Banana, Maize, Beans, Coffee, Fish, Livestock products	maize, Beans, cassava, coffee, charcoal	maize, bananas, beans coffee electricity poles
	What are the main challenges reported by the District Local Government?			
	Lack of knowledge on soil improvement	Lack of knowledge on soil improvement techniques	Lack of access to markets	Low crop yields
	Lack of access to markets	Lack of farm inputs	Lack of knowledge on soil improvement techniques	Lack of knowledge on soil improvement techniques

	Lack of farm inputs	Low crop yields	Increased insurgencies of pests and diseases	
	Low crop yields	Lack of access to markets	Lack of farm inputs	
	Increased insurgencies of pests and diseases	Increased insurgencies of pests and diseases	Low crop yields	
	Low Farm incomes	Low Farm incomes	Low Farm incomes	
	What is the average Livelihoods breakdown in the sampled area?			
cash crops	60%	60%	60%	75%
trees	15%	20%	2%	20%
apiculture	0	20%	38%	5%
livestock	25%	0	0	0
fisheries	1%	0	0	0
	Do you have any recent production reports or statistics for the district that could be shared			
	yes=1,	No=1	No=1,	yes=1,
	Are there any targets and progress plans for these reports or statistics for the district			
	yes=1,	No=1,	No=1,	yes=1,
C. Department of Trade, Industry and Local Economic Development				
	What are the main economic opportunities in the District?			
	Agriculture, horticulture, beef & dairy, Apiculture, tourism, finance	Agriculture &livestock farming	Commercial agriculture, Trading	Agro-forestry, Tourism, Sandmining & stone mining
	What are the main economic opportunities in the District Local Government? (Rank in order of decreasing severity i.e., from most to least severe in the list).			
	Fewer business opportunities	Bad terrain hindering transport of crop yields	Lack of business infrastructure e.g. roads, internet, legal and banking?	Fewer business opportunities

	Lack of information on available business opportunities	Lack of business infrastructure e.g. roads, internet, legal and banking	Lack of information on available business opportunities	Lack of starter capital
	Lack of starter capital	Lack of starter capital	Fewer business opportunities	Lack of business infrastructure e.g. roads, internet, legal and banking
		Low returns on investments	Low returns on investments	
	Lack of business infrastructure e.g. roads, internet, legal and banking	Fewer business opportunities	O Bad terrain hindering transport of crop yields	
		Lack of information on available business opportunities	Lack of starter capital	
	Do you have any recent business reports or statistics for the district that could be shared			
	yes=1,	, No=1	yes=1,	yes=1,
	Are there any targets and progress plans for these reports or statistics for the district			
	yes=1,	, No=1	yes=1,	yes=1,
D. Department District Community Development				
	What are the main challenges reported by the District Local Government? (Rank in order of decreasing severity i.e., from most to least severe in the list).			
	O Increased domestic violence	Increased early pregnancies	Increased early marriages	0
	Increased early marriages	O Increasing number of thefts /crimes	Increasing number of thefts / crimes	0
	Access to education	Gender / youth issues	Access to healthcare	0
	Access to healthcare	Gender / youth issues	O Increased early pregnancies	0
	Gender / youth issues	Increased domestic violence	Gender / youth issues	0

	Increased early pregnancies	Increased early marriages	Increased early marriages	0
	land conflicts		Increased poverty levels	0
			Increased domestic violence	0
	What are the main community development projects in this district?			
	NAADS,HEIFER,FE FOC,FISHERIES			VSLA & Function Adult Program (FAP)
	Do you have any recent Community development reports or statistics for the district that could be shared?			
	yes=1,	, No=1	yes=1,	yes=1,
	Are there any targets and progress plans for these reports or statistics for the district			
	yes=1,	, No=1	yes=1,	yes=1,
E: Department of Education				
	What are the main focus areas of the district?			
	Quality social service	Provide education	addressing poverty & domestic violence	UPE
	What are the main challenges reported by the District Local Government? (Rank in order of decreasing severity i.e. from most to least severe in the list).			
	Long distances of commuting from School-Home	Lack of pedagogical and scholastic materials	Long distances of commuting from School-Home	Increased rate of school dropouts
	Unavailability of school feeding programs	Low literacy levels	Lack of pedagogical and scholastic materials	Lack of pedagogical and scholastic materials
	Lack of pedagogical and scholastic materials	Child Labor	Lack of access to power / water at schools?	Child Labor
	Lack of school infrastructure	Increased rate of school dropouts	Lack of school infrastructure?	inadequate teachers
	Child Labor	Lack of school infrastructure?	Low literacy levels	
	Increased rate of school dropouts	Lack of access to power / water at schools?	Unavailability of school feeding programs	

	Low literacy levels	Unavailability of school feeding programs	Child Labor	
	Lack of access to power / water at schools	Long distances of commuting from School-Home	Increased rate of school dropouts	
	Do you have any recent Education reports or statistics for the district that could be shared			
	yes=1,	yes=1,	yes=1,	yes=1,
	What % complete primary school?			
	60%		80%	87%
	What % complete High school?			
	30%		60%	69%
	What % complete tertiary education?			
	10%		15%	60%
	Are there any targets and progress plans for these reports or statistics for the district			
	yes=1,	Yes=1,	Yes=1,	Yes=1,

9. Conclusion

From the analysis of the lite Social and Environmental study conducted by NFC with neighboring communities it is clear from a social perspective that the needs for livelihoods development are the main priority, to continue to build agency and avoid overdependency on the company. This includes a focus on youth and gender; financial inclusion and access to loans to start small businesses. Access to micro insurance to help communities adapt to health or business shocks is an area to be further investigated. The carbon project will support additional revenue streams for Out-grower households, through the Out-grower Associations and specific measures will be needed to ensure the inclusion of youth and gender in realizing these benefits. In the absence of piped water, access to clean water links directly to the health of the environment.

From an environmental perspective, climate change is a growing priority that has many related impacts, such as on biodiversity and the state of riparian zones, but also on access to drinking water and the ability of farmers to sustain themselves. The carbon project will likely have a positive impact in that it will promote tree growing, but a riparian strategy will be needed to mitigate the risks of tree growing in wetlands, or potential land conflicts that could arise. The carbon landscape project aligns with the objectives documented of local government and the National government goal to increase forest cover.